Chapter 4

Estimation, Testing, and Prediction in the Simple Linear Regression Model

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MGMTMSA 400

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Prediction under the Simple Linear Regression Model

Prediction and the Modeling Goal

- A **prediction rule** uses a value of X to predict an outcome Y that we have not observed

$$\hat{Y} = f(X) = b_0 + b_1 X$$

- In the video card example from Chapter 3, X is the performance score and Y is the natural log of the listed price in dollars

$$\widehat{\log(\text{price})} = \underbrace{4.897 + 0.00753\, X}_{\text{Prediction Rule}}$$

- A useful prediction also needs a measure of uncertainty

Prediction and the Modeling Goal

- To assess prediction accuracy, we need a **probability model** for how outcomes vary
  - A probability model describes how population data are generated, using unknown parameters
  - Recall the chip example in Chapter 2: we assumed normally distributed performance scores and used a sample to estimate the mean and standard deviation
- For the video card example, we use the **simple linear regression (SLR) model**:

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

- The model separates Y into three components:
  - A constant (β_0)
  - A component that changes linearly with X ($\beta_1 X$)
  - A random deviation that we cannot predict from X (ε)

Conditional Distributions and the Regression Model

- A **conditional distribution** describes how Y varies when X takes a particular value
  - For video cards with a performance score of 200, what is the distribution of their log prices?
- In the normal SLR model, errors are independent across observations and independent of X, with $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$

$$\mathbb{E}[Y \mid X = x] = \beta_0 + \beta_1 x$$
$$\text{Var}(Y \mid X = x) = \sigma^2$$

- Therefore, the conditional distribution is

$$Y \mid X = x \sim \mathcal{N}(\beta_0 + \beta_1 x, \sigma^2)$$

  - This assumption does not require X to have a normal distribution

Conditional vs. Unconditional Distributions

- When X varies across the population, the unconditional variance of Y includes two sources of variation:

$$\begin{aligned}\text{Var}(Y) &= \text{Var}(\beta_0 + \beta_1 X + \varepsilon) \\ &= \beta_1^2 \,\text{Var}(X) + \sigma^2\end{aligned}$$

  - Variation associated with different values of X: $\beta_1^2 \,\text{Var}(X)$
  - Variation remaining at any given X: σ^2
- Knowing X removes the first source of uncertainty

$$\text{Var}(Y \mid X = x) = \sigma^2 \leq \text{Var}(Y)$$

- Within this model, knowing X reduces uncertainty whenever $\beta_1 \neq 0$ and X varies across observations

The Simple Linear Regression Model vs. Linear Prediction

- SLR:

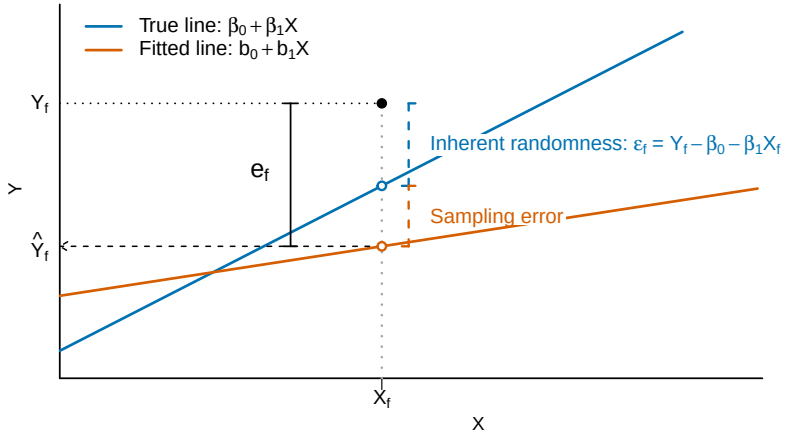
$$Y = \beta_0 + \beta_1 X + \varepsilon$$

- Linear prediction:

$$\hat{Y} = b_0 + b_1 X$$

- If we knew β_0, β_1, and σ, we could use the conditional distribution directly. In practice, we estimate β_0 and β_1 using the least-squares estimates b_0 and b_1
- Therefore, the following factors affect the accuracy of our predictions:
  - The error term ε
  - The difference between estimates $\{b_0, b_1\}$ and true parameters $\{\beta_0, \beta_1\}$

Visualization: Sources of Prediction Errors



- At X_f, the data point we will observe is (X_f, Y_f), while our prediction is $\hat{Y}_f$
- The total prediction error is $e_f = Y_f - \hat{Y}_f = \varepsilon_f + (\beta_0 - b_0 + (\beta_1 - b_1)X_f)$

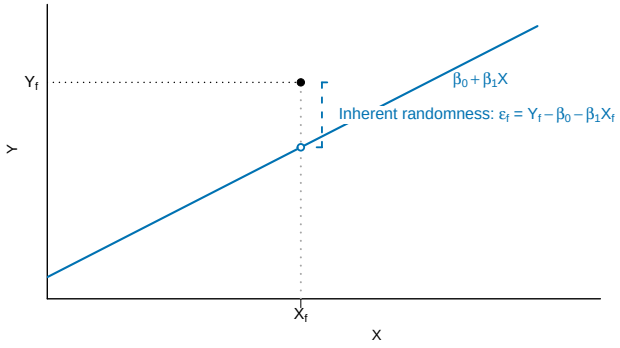
Outline of This Module

- How is prediction accuracy affected by the error term ε?
  - What are the properties of ε in our probability model?
  - How can we estimate the variability of ε from sample data?
- How is prediction accuracy affected by the estimates b_0, b_1?
  - How and why might they differ from the true parameters β_0, β_1?
- Putting it all together: How do we construct a prediction interval for a new observation with input value X_f?

The Error Term

Properties of the Error Term

- To focus on the error term, we **temporarily** assume that the true parameters β_0 and β_1 are known
- The error term captures variation in Y that is not explained by X
  - For cards with the same performance score, $\mathbb{E}[Y|X]$ is the average log price. Individual cards may differ from this average



Properties of the Error Term

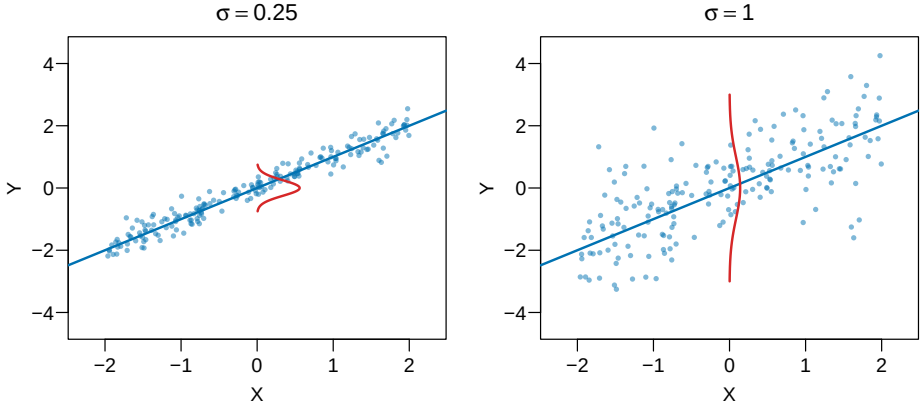
- In the normal SLR model, we assume that

$$\varepsilon_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2), \qquad \varepsilon_i \text{ is independent of } X_i$$

- The error term has mean zero at every value of X: knowing X does not help us predict the average error
- A normal distribution can be a useful approximation when many small, independent influences contribute to the error
- These are modeling assumptions and can be relaxed
  - In Chapter 5, we will allow the error variance to vary with X. The causal inference course will examine what happens when X and ε are related

The Simple Linear Regression Model

- σ determines the dispersion of points about the true line
- The blue line is the true conditional mean; each red curve shows the distribution of Y at $X = 0$



Estimation of σ

- In practice, the true line is unknown. We estimate it by least squares and use the residuals e_i to estimate σ
- An unbiased estimator of the error variance $\sigma^2 = \mathbb{E}[\varepsilon_i^2]$ is

$$s^2 = \frac{\text{SSE}}{N-2} = \frac{\sum_{i=1}^{N} e_i^2}{N-2}$$

  - Fitting an intercept and a slope uses two degrees of freedom, leaving $N-2$
- Taking the square root gives an estimate of the error standard deviation:

$$s = \sqrt{\frac{\text{SSE}}{N-2}} = \text{standard error of the regression}$$

  - s estimates the standard deviation σ of the model errors

Estimation of σ in R

```
summary(out_vc)
##
## Call:
## lm(formula = log(price_usd) ~ performance, data = vc)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.00693 -0.29504 -0.09091  0.26087  2.58868
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 4.8968245  0.0800635   61.16   <2e-16 ***
## performance 0.0075311  0.0004826   15.61   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4855 on 123 degrees of freedom
## Multiple R-squared:  0.6644, Adjusted R-squared:  0.6617
## F-statistic: 243.6 on 1 and 123 DF,  p-value: < 2.2e-16
```

- s is the "Residual standard error": 0.4855 log-price units

Statistical Inference on the Estimates

Estimation in the Simple Linear Regression Model

- Recall the simple linear regression model:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i \qquad (i = 1, ..., N)$$

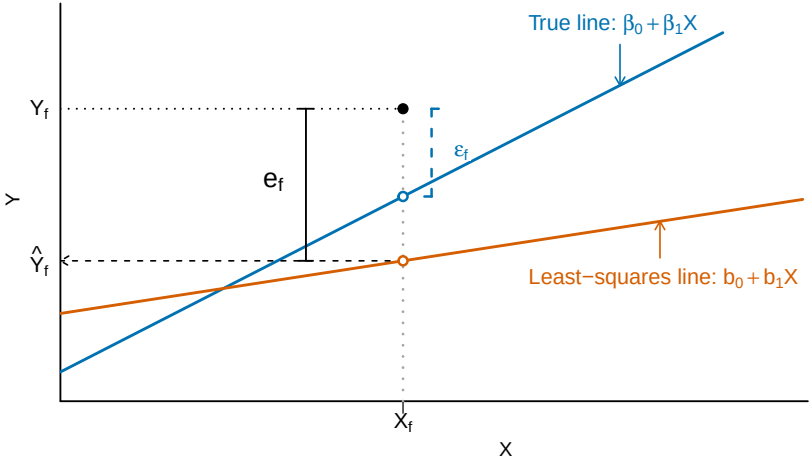
- We use **least squares** to estimate β_0 and β_1

$$b_1 = \frac{\sum_{i=1}^{N}(X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^{N}(X_i - \bar{X})^2}$$

$$b_0 = \bar{Y} - b_1\bar{X}$$

Estimation in the Simple Linear Regression Model

Distinguish the **population parameters** (β_0, β_1) from their **sample estimates** (b_0, b_1), and the model error ε_i from the residual e_i



Using Simulation to Understand the Sampling Distribution of b_1

- Different samples give different coefficient estimates; we use simulation to study the **sampling distribution** of b_1
- Keep the 125 observed video card performance scores X_i fixed and generate log prices from:

$$Y_i = 6 + 0.01X_i + \varepsilon_i, \quad \varepsilon_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, 0.25^2)$$

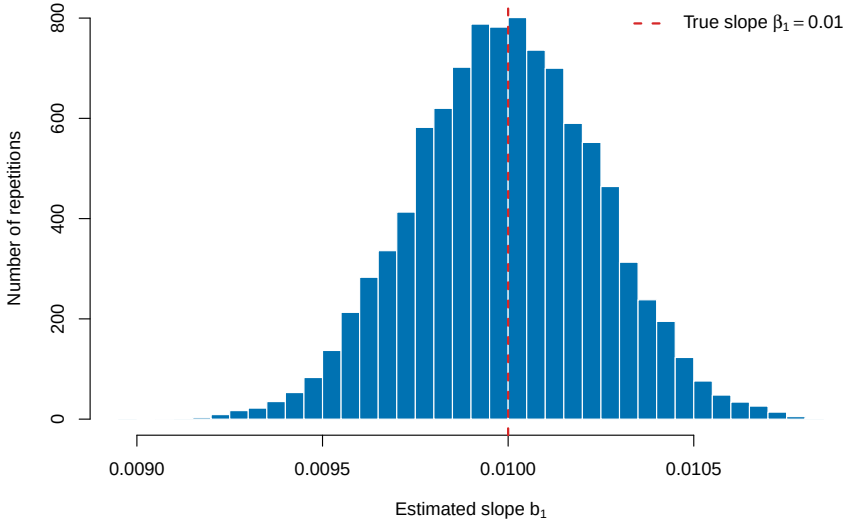
- These parameters are chosen for the simulation, so we know the true slope is $\beta_1 = 0.01$
- Repeat the following steps 10,000 times:
  - Simulate new error terms ε
  - Generate a new log price Y_i for each performance score
  - Refit the regression and record the new estimate of b_1

Using Simulation to Understand the Sampling Distribution of b_1

- Each repetition generates 125 log prices and fits a new line

```
set.seed(1234)
x <- vc$performance
beta0 <- 6; beta1 <- 0.01; sigma <- 0.25
nsample <- 10000
bsim <- double(nsample)
for (i in seq_len(nsample)) {
  y <- beta0 + beta1 * x + rnorm(length(x), sd = sigma)
  bsim[i] <- coef(lm(y ~ x))[2]
}
```

Using Simulation to Understand the Sampling Distribution of b_1



Sampling Distribution of b_1

- The simulated slopes form a bell-shaped distribution centered near the true slope $\beta_1 = 0.01$
- Substituting the model for Y_i in the slope formula gives:

$$b_1 = \frac{\sum_i (X_i - \bar{X})\left[(\beta_0 + \beta_1 X_i + \varepsilon_i) - (\beta_0 + \beta_1 \bar{X} + \bar{\varepsilon})\right]}{\sum_i (X_i - \bar{X})^2}$$

$$\implies b_1 = \beta_1 + \sum_i \frac{X_i - \bar{X}}{\sum_j (X_j - \bar{X})^2}\,\varepsilon_i$$

- With the X_i values fixed, b_1 is a constant plus a linear combination of normal errors, so it is normally distributed

Sampling Distribution of b_1

- Under our normal-error model, holding the observed X_i values fixed:

$$b_1 \sim \mathcal{N}(\beta_1, \sigma^2_{b_1})$$

  - The mean of b_1 is β_1, so b_1 is an unbiased estimator
  - $\sigma^2_{b_1}$ measures the variation in b_1 across repeated samples
- A smaller variance means the estimates are more concentrated around the true slope

Sampling Distribution of b_1

- Since

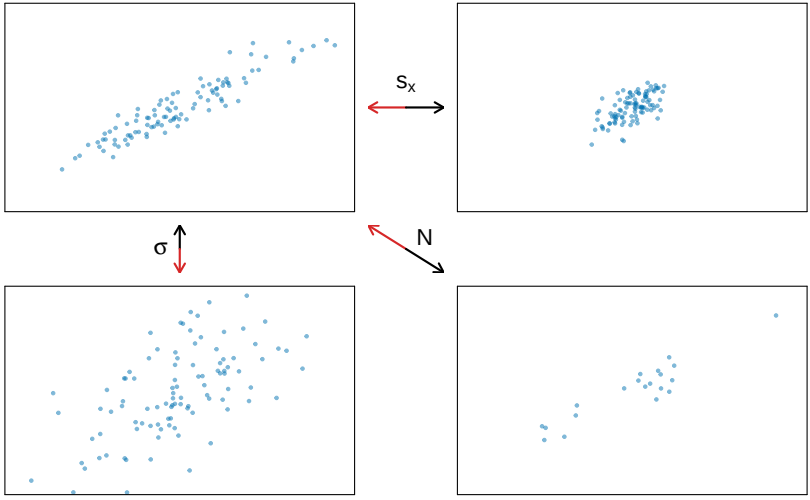
$$b_1 = \beta_1 + \sum_i \frac{(X_i - \bar{X})}{\sum_j (X_j - \bar{X})^2} \varepsilon_i$$

we can also derive

$$\begin{aligned} \text{Var}(b_1) &= \text{Var}\left(\sum_{i=1}^{N} \frac{(X_i - \bar{X})}{\sum_{j=1}^{N} (X_j - \bar{X})^2} \varepsilon_i \right) = \sum_{i=1}^{N} \frac{(X_i - \bar{X})^2}{[\sum_{j=1}^{N} (X_j - \bar{X})^2]^2} \text{Var}(\varepsilon_i) \\ &= \frac{\sigma^2}{\sum_{i=1}^{N} (X_i - \bar{X})^2} = \frac{\sigma^2}{(N-1)s_x^2} \end{aligned}$$

- What is the intuition behind the formula $\sigma^2_{b_1}$?

Sampling Distribution of b_1



- Note: The red side of arrows indicates an increase in the parameter

Understanding Standard Errors of b_1

- We do not know the error standard deviation σ, so we estimate it with the residual standard deviation s
- Replacing σ with s gives the **estimated standard error** of the slope:

$$s_{b_1} = \sqrt{\frac{s^2}{(N-1)s_x^2}}$$

- s_{b_1} estimates how much the slope estimate would vary across repeated samples
- A smaller standard error means a more precise estimate of β_1

Standard Errors of b_1 and b_0 in R

- Return to the regression fitted to the 125 observed video card prices

```
coef(summary(out_vc))
##                Estimate   Std. Error  t value     Pr(>|t|)
## (Intercept) 4.896824543 0.0800634590 61.16179 6.180507e-94
## performance 0.007531139 0.0004825651 15.60647 5.987307e-31
```

- The estimated standard error of the intercept is $s_{b_0} = 0.08006$
- The estimated standard error of the performance coefficient is $s_{b_1} = 0.000483$

Confidence Intervals

- A confidence interval gives a range of plausible values for a population coefficient
- A $(1-\alpha) \times 100\%$ confidence interval for the slope is:

$$b_1 \pm t^{N-2}_{1-\alpha/2} s_{b_1}$$

- Use the t-distribution with $N-2$ degrees of freedom under the normal-error model
- In R, `confint()` gives 95% confidence intervals by default

```
confint(out_vc)
##                    2.5 %      97.5 %
## (Intercept) 4.738343835 5.055305251
## performance 0.006575931 0.008486347
```

Hypothesis Testing

- We are often interested in a specific value of the slope parameter, β_1
- A common question is whether the data provide evidence of a linear relationship between *X* and *Y*
  - Are years of education associated with income?
  - What are the null and alternative hypotheses for this test?
- We use **one-tailed tests** when the question concerns a particular direction or threshold
  - Example: Does the association exceed a business target?

Hypothesis Testing: Choice of α

- Choosing the significance level α involves a trade-off between two types of errors:
  - **Type I error:** Rejecting the null hypothesis when it is actually true
  - **Type II error:** Failing to reject the null hypothesis when it is actually false
- A common choice is $\alpha = 0.05$, but there is no universally best value
  - If the cost of a Type I error is high (e.g., approving an ineffective drug), we may choose a smaller α
  - With the same data and test, a smaller α also makes a real effect harder to detect

Hypothesis Testing: *t*-Statistic and *P*-value

- Calculate the *t*-statistic using the value β_1^* specified by the null hypothesis:

$$t = \frac{b_1 - \beta_1^*}{s_{b_1}} = \frac{\text{estimate} - \text{hypothesized value}}{\text{standard error}}$$

- Use the *t*-distribution with $N - 2$ degrees of freedom to calculate the *p*-value
  - Use the upper tail for $H_A : \beta_1 > \beta_1^*$, the lower tail for $H_A : \beta_1 < \beta_1^*$, and both tails for $H_A : \beta_1 \neq \beta_1^*$
- Finally, draw a conclusion:

$$p < \alpha \longrightarrow \textbf{reject} \text{ the null hypothesis}$$
$$p \geq \alpha \longrightarrow \textbf{fail to reject} \text{ the null hypothesis}$$

Example: Testing a Video Card Pricing Target

- Is a 1-point higher performance score associated with **more than a 1% higher typical listed price**?
- In our log-price model, a 1-point increase in X is associated with approximately a $100\beta_1\%$ change in typical listed price, so we test the slope against 0.01
- Use a **one-tailed test** with the upper tail at the 5% significance level:

$$H_0 : \beta_1 \leq 0.01 \qquad H_A : \beta_1 > 0.01$$

- This tests an association; it does not establish the effect of changing a card's performance

Example: Video Card Test Results in R

```
target <- 0.01
b1 <- coef(out_vc)[["performance"]]
se_b1 <- coef(summary(out_vc))["performance", "Std. Error"]
t_stat <- (b1 - target) / se_b1
df <- df.residual(out_vc)
p_value <- pt(t_stat, df = df, lower.tail = FALSE)
round(c(slope = b1, SE = se_b1, t = t_stat, df = df), 6)
##      slope         SE          t         df
##   0.007531   0.000483  -5.116120 123.000000
sprintf("One-tailed p-value: %.7f", p_value)
## [1] "One-tailed p-value: 0.9999994"
```

- The estimated slope is 0.007531, below the threshold 0.01; the one-tailed test gives $t = -5.12$ and $p > 0.99$
- Since $p > 0.05$, we **fail to reject** H_0: the data do not provide evidence that a 1-point higher score is associated with more than a 1% higher typical listed price
- The fitted model associates a 1-point higher performance score with approximately a 0.75% higher typical listed price

Construction of Prediction Intervals

Prediction

- Recall the normal SLR model:

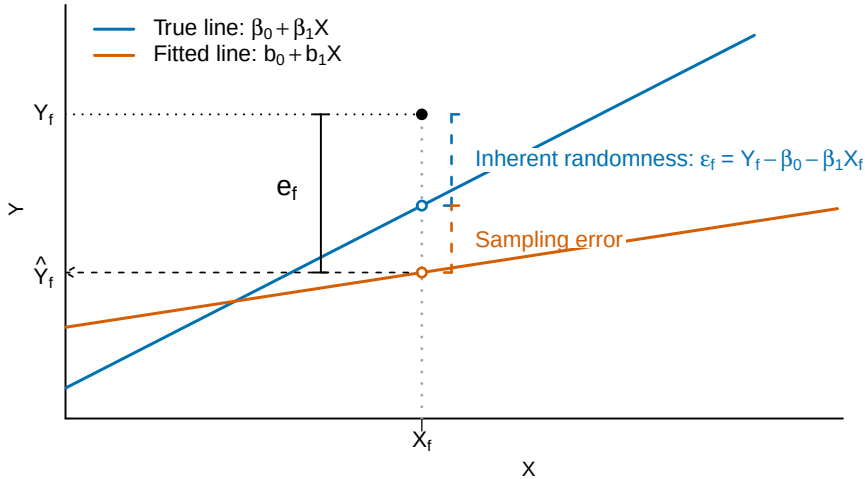
$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i, \qquad \varepsilon_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2)$$

- We want to predict a new outcome Y_f, given its input X_f and our sample data
- Our point prediction uses the estimated coefficients:

$$\hat{Y}_f = b_0 + b_1 X_f$$

- The **prediction error** is $e_f = Y_f - \hat{Y}_f$

Visualization of Prediction Error (Revisit)



Prediction Error: Two Sources

- The prediction error combines two sources of uncertainty:
  - Inherent randomness: the new outcome differs from its true conditional mean
  - Sampling error: our fitted line differs from the true line

$$e_f = Y_f - \hat{Y}_f = (Y_f - \beta_0 - \beta_1 X_f) + (\beta_0 + \beta_1 X_f - \hat{Y}_f)$$
$$= \underbrace{\varepsilon_f}_{\text{Inherent randomness}} + \underbrace{\left[(\beta_0 - b_0) + (\beta_1 - b_1)X_f\right]}_{\text{Sampling error}}$$

- More data can reduce uncertainty about the fitted line, but the new observation still has its own error ε_f

Prediction Interval

- Under the normal SLR model, a $100(1-\alpha)\%$ prediction interval for one new outcome at X_f is

$$b_0 + b_1 X_f \pm t^{N-2}_{1-\alpha/2} s \sqrt{1 + \frac{1}{N} + \frac{(X_f - \bar{X})^2}{(N-1)s_x^2}}$$

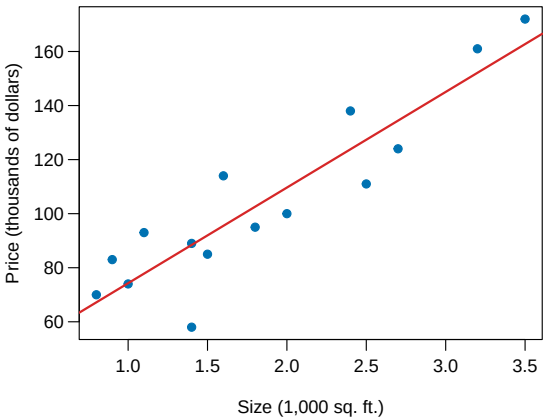
- In R, use `predict()` with `interval="prediction"`
- For a video card with performance score 200, calculate a 95% interval and use `exp()` to convert its endpoints from log-price units to dollars

```
log_interval <- predict(out_vc, newdata=data.frame(performance=200),
                        interval="prediction", level=0.95)
round(exp(log_interval), 2)
##      fit    lwr     upr
## 1 603.68 229.61 1587.16
```

- The predicted typical listed price is about \$604, with a 95% prediction interval from about \$230 to \$1,587

R^2 and Prediction Accuracy

- Consider the relationship between house size and price in the `housepr` dataset



- $R^2 = 0.83$
- $s \approx \$14,138$
- Average price: about \$104,467
- At the average size, the 95% prediction interval is approximately $[\$72,921,\ \$136,013]$

- A high R^2 does not guarantee a narrow prediction interval: is this precise enough for a pricing decision?

Appendix

Glossary of Symbols

Model parameters and prediction notation:

- β_0 = true line intercept
- β_1 = true slope
- σ or σ_ε = error standard deviation
- ε = model error: the deviation from the true regression line
- X_f = input value for a new observation
- Y_f = new outcome to be predicted
- ε_f = prediction error when predicting from the true line
- e_f = prediction error when predicting from the fitted line

Important Equations

Two versions of the normal simple linear regression model:

- As a regression equation:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i \quad \text{with} \quad \varepsilon_i \sim \text{iid}\, \mathcal{N}(0, \sigma_\varepsilon^2)$$

- As a conditional distribution:

$$Y|X = x \sim \mathcal{N}(\beta_0 + \beta_1 x,\, \sigma^2)$$

Important Equations

- Estimated error standard deviation (standard error of the regression):

$$s = \sqrt{\frac{\text{SSE}}{N-2}}$$

- Root Mean Squared Error:

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(Y_i - \hat{Y}_i\right)^2}$$

Important Equations

- Estimated standard error of the slope:

$$s_{b_1} = \sqrt{\frac{s^2}{(N-1)s_x^2}}$$

- Confidence interval:

$$100(1-\alpha)\%\text{ C.I.}: \quad b \pm t_{1-\alpha/2}^{N-2} s_b$$

- Rejection region for a two-tailed t-test

$$|t| = \left| \frac{b_1 - \beta_1^*}{s_{b_1}} \right| > t_{1-\alpha/2}^{N-2}$$

- Two-tailed p-value under the null hypothesis:

$$p = \Pr\big(|T| \geq |t_{\text{obs}}|\big), \qquad T \sim t_{N-2}$$

Important Equations

- Definition of prediction error:

$$e_f = Y_f - \hat{Y}_f = Y_f - b_0 - b_1 X_f$$

- For a new observation independent of the sample, treating the X values as fixed:

$$\text{Var}(e_f) = \sigma^2 + \text{Var}(\hat{Y}_f)$$

Glossary of R Commands

- `confint(out_lm)` Compute confidence intervals for the coefficients of the fitted model `out_lm`